



Brain-like chips and future technology innovations

IBM, a technology company, has announced that its prototype 'brain-like' chip can potentially enhance energy efficiency in artificial intelligence (AI). The new prototype chip mimics neural connections in the human brain, enabling complex tasks in low-power environments like cars and phones, and helping cloud providers reduce energy costs and environmental impact. Unlike traditional digital chips which only store data as 0s and 1s, this chip uses analogue memristors capable of storing varied values, replicating brain functions and forming networks akin to a biological brain. Despite the complex development process and high material costs, its integration enhances energy efficiency while maintaining digital features, facilitating its incorporation into existing AI systems and signalling the near emergence of brain-like chips.

Collecting clean water from fog

Researchers at ETH Zurich have developed a technique to harvest and simultaneously purify water from fog. The technology requires little to no maintenance once installed,



A mesh coated with titanium dioxide harvests and cleans water from fog (Credit: ETH Zurich/Adobe Stock/Montage)

utilising a brief, regular UV exposure to rejuvenate the catalyst. A mere half-hour of sunlight can reactivate the titanium oxide for a full day due to its photocatalytic memory, remaining effective for long periods even without light, which is vital in fog-prone areas. Lab and Zurich pilot plant tests showed the fog collector captured 8% of water from synthetic fog and removed 94% of organic compounds, including fine diesel particles and bisphenol A. Besides extracting drinking water from fog, this technology holds the potential for reclaiming water from cooling towers. In cooling towers, steam dissipates into the air.

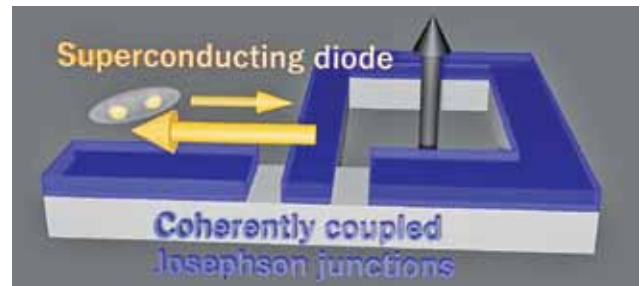
A recyclable organo-ionic electrolyte that can be liquefied

Engineers from the molecular foundry and the energy storage division at Lawrence Berkeley National Laboratory have crafted an electrolyte material for solid-state batteries. This

advancement permits the cathodes to be reused with up to 90% of their original capacity. The team has developed a more recyclable lithium-ion battery using a newly formulated electrolyte infused with organic-based zwitterionic polymers and paired with various material cathodes. This innovation allows the electrolyte to melt at 100°C, simplifying battery deconstruction and recycling. Notably, the electrolyte maintains solid form and high conductivity at standard operating temperatures, even up to 45°C. Moreover, the disassembled battery's cathode retains 90% of its initial capacity, streamlining the recycling process, as noted by the researchers.

The superconducting diode effect in coupled Josephson junctions

Researchers from RIKEN and other institutes in Japan and the US have recently detected the SC diode effect in a novel device comprising two interconnected Josephson junctions (JJs). The team managed to generate the superconducting



The superconducting diode effect (Credit: Matsuo et al)

(SC) diode effect by modulating the phase of these interconnected JJs, offering valuable insights into achieving this effect in devices reliant on coupled JJs and enhancing the understanding of the fundamental physics involved. This SC diode effect, which occurs when SC devices lack both time-reversal and spatial-inversion symmetries, holds significant potential for energy-efficient rectification in forthcoming SC circuits. The results suggest that manipulating the phase of the connected JJs can breach these symmetries, paving the way for the possible emergence of other distinct SC phenomena in coupled JJs when these symmetries are disrupted.

Improving EV battery longevity with real world data

Researchers at Stanford University have developed an algorithm to demonstrate the gap between controlled laboratory testing and actual road experience. Using Audi's field data, researchers analysed the battery pack's electrical resistance over a year to assess its energy and power. They determined resistance values from 529 acceleration and 392 braking events, and impedance from 53 charging events.